Homework 3 – Regression

1 Question 1

Below, I simulate a rather quite annoying set of regression data.

```
set.seed(0510)
n <- 250
dat.q1 <- tibble(x1 = rnorm(n, mean = 50, sd = 10),
                 x2 = factor(sample(c("A", "B"), n, replace = TRUE)),
                 x3 = factor(sample(c("Control", "Treat"), n, replace = TRUE)),
                 x4 = factor(sample(c("Low", "High"), n, replace = TRUE), levels=c("Low", "High")),
                 x5 = rnorm(n, mean = 0, sd = 1),
                 x6 = factor(sample(c("Group1", "Group2"), n, replace = TRUE)),
                 x7 = rnorm(n, mean = 10, sd = 2),
                 x8 = rnorm(n, mean = 10, sd = 2)) |>
        mutate(y = 5 +                          # Intercept
                   # Quantiative x1
                   2 * x1 +
                   # Categorical x2
                   (x2 == "B") * 10 +
                   # Categorical x Categorical (x3*x4)
                   (x3 == "Treat") * 5 + (x4 == "High") * 2 +
                   (x3 == "Treat" & x4 == "High") * 15 +
                   # Quantitative x Categorical  (x5*x6)
                   3 * x5 + (x6 == "Group2") * 4 +
                   (x5 * (x6 == "Group2")) * 8 +
                   # Quantitative x Quantitative (x7*x8)
                   1.5 * x7 + 0.5 * x8 +
                   0.8 * (x7 * x8) +
                   # Error
                   rnorm(n, mean = 0, sd = 15))
```

1.1 Part a

Fit the regression model.

$$
\begin{aligned}
Y_i = &\beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \beta_3 I(x_{3i} = \text{Treat}) + \beta_4 I(x_{4i} = \text{High}) + \\
&\beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\
&\beta_9 (I(x_{3i} = \text{Treat}) \times I(x_{4i} = \text{High})) + && \text{(Categorical} \times \text{Categorical)} \\
&\beta_{10} (x_{5i} \times I(x_{6i} = \text{Group2})) + && \text{(Continuous} \times \text{Categorical)} \\
&\beta_{11} (x_{7i} \times x_{8i}) + \epsilon_i && \text{(Continuous} \times \text{Continuous)}
\end{aligned}
$$

Solution: First, we fit the regression model using `lm()`.

```
mod.q1 <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.q1)
```

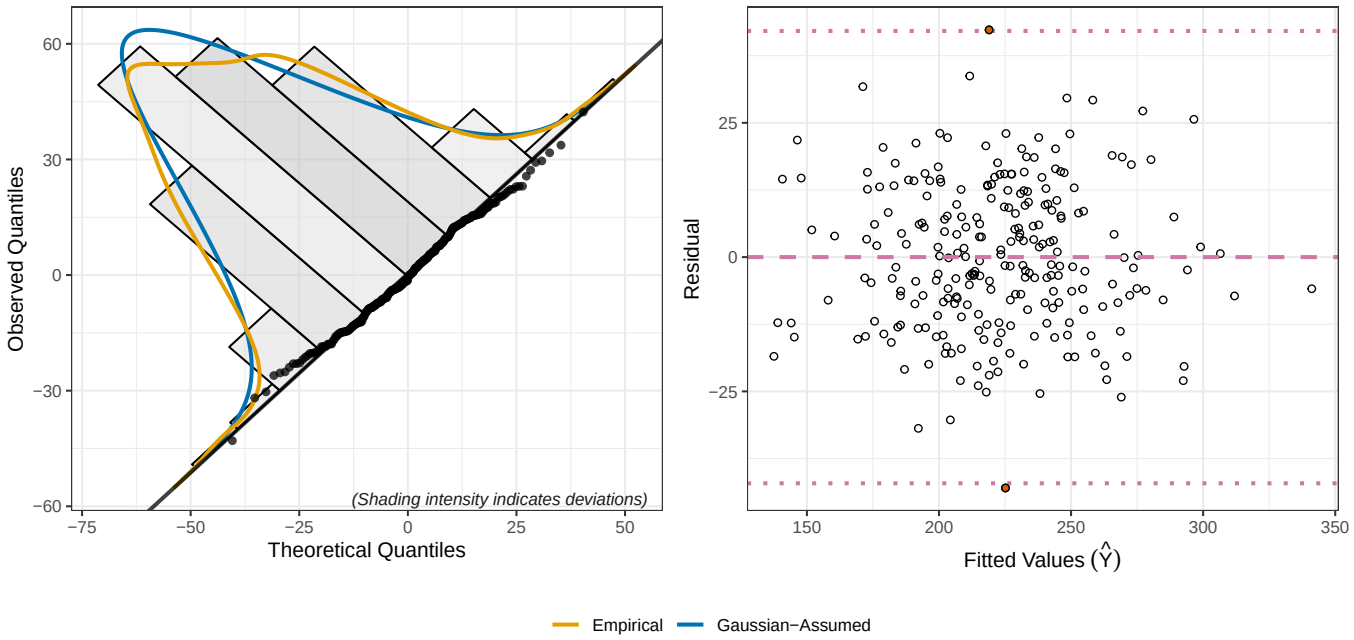
1.2 Part b

Conduct a residual and outlier analysis.

Residual Analysis

```r
plotResiduals(mod.q1)
```

Figure 1: Residual diagnostics for the Question 1 model.

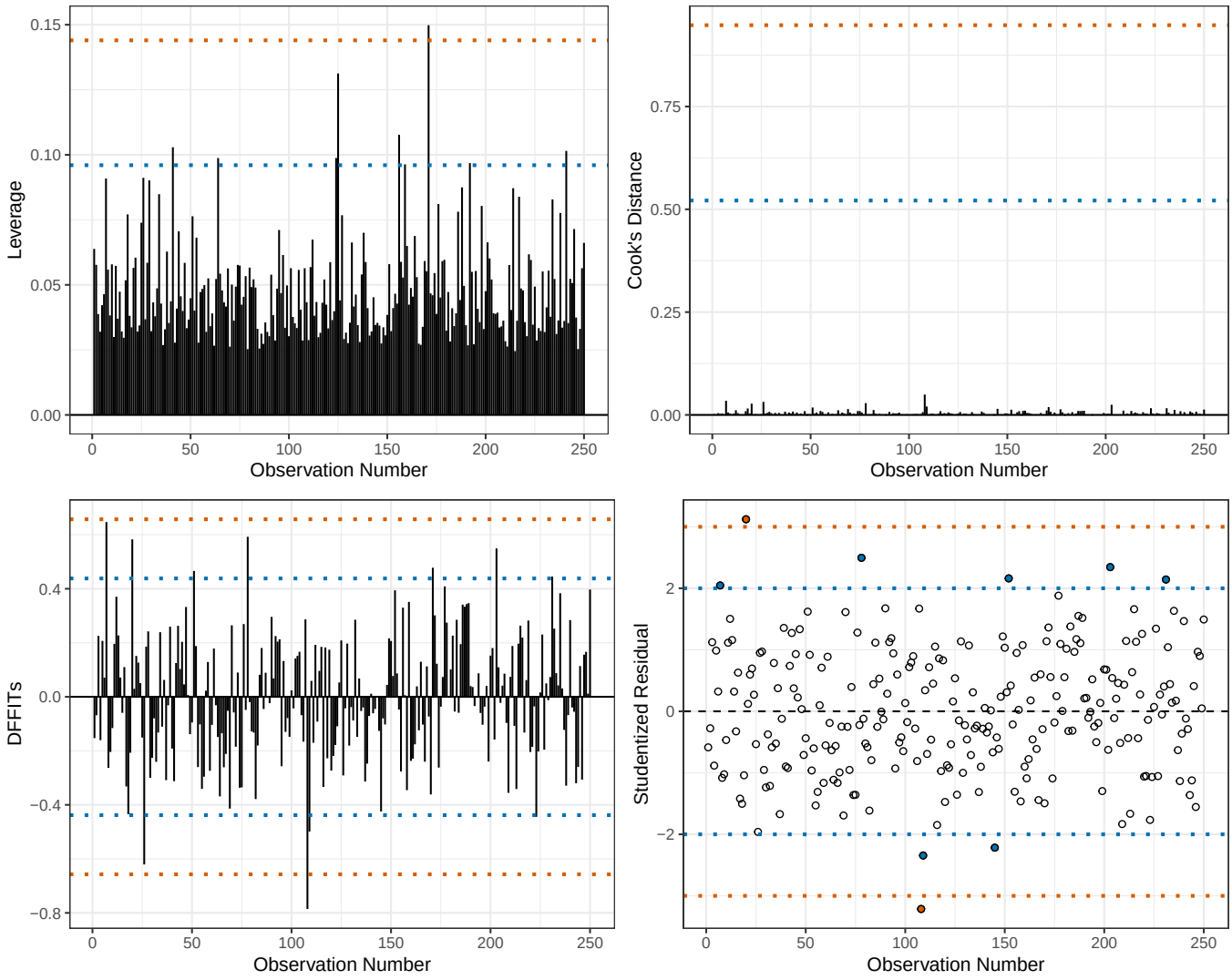


The residuals follow a Gaussian distribution and appear balanced around 0. The residuals are independent via the simulated data.

Outlier Analysis

```r
plotInfluence(mod.q1)
```

Figure 2: Residual diagnostics for the Question 1 model.



There are very few outliers, and most only have a small effect on the regression model.

1.3 Part c

Interpret the intercept, the coefficients on x_1 and x_2.

$\hat{\beta}_0$: The expected value of $\hat{Y} = 8.9653$ when the quantitative variables x_1, x_5, x_7, x_8 are 0 and when the categorical variables $x_2 =$ "A", $x_3 =$ "$Control$", $x_4 =$ "Low", and $x_6 =$ "$Group1$".

$\hat{\beta}_1$: The expected change in $\hat{Y} = 2.13546$ for a one-unit increase in x_1, all other variables constant.

$\hat{\beta}_2$: The expected change in $\hat{Y} = 10.02851$ for a one-unit increase in x_2, all other variables constant.

1.4 Part d

Untangle and interpret the $x_3 * x_4$ interaction using algebra. Use Table 1 and Figure 3 for context.

```
library(emmeans)
emm.q1d <- emmeans(mod.q1, ~x3*x4) |> # Don't average across levels: use |x2*x6
            data.frame() |>
            rename("Mean"  = "emmean",
```

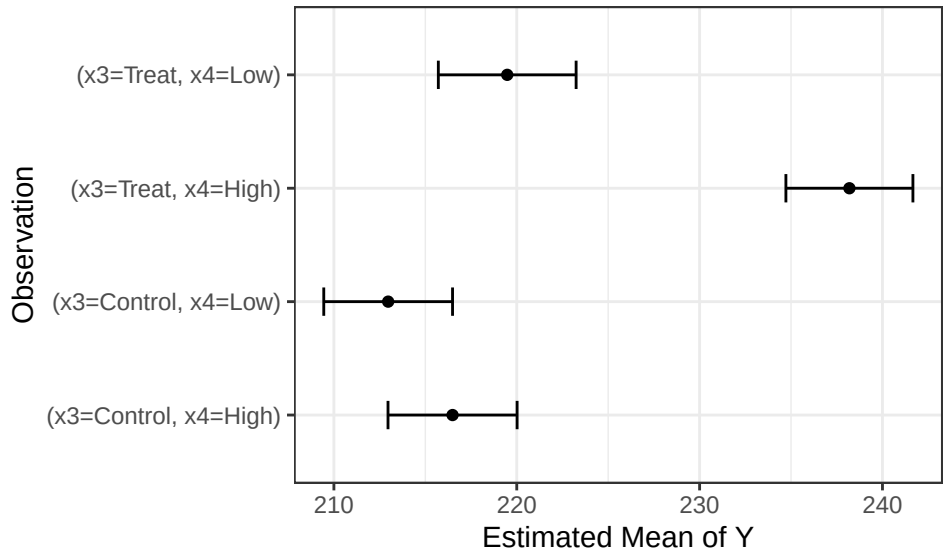
```
                    "Lower" = "lower.CL",
                    "Upper" = "upper.CL")
```

Table 1: Estimated marginal means for the model in Question 1.

x3	x4	Mean	SE	df	Lower	Upper
Control	Low	212.9676	1.7871	238	209.4470	216.4882
Treat	Low	219.4792	1.9123	238	215.7120	223.2464
Control	High	216.4894	1.7916	238	212.9601	220.0188
Treat	High	238.1930	1.7631	238	234.7198	241.6662

```
emm.q1d <- emm.q1d |>
  mutate(Observation = paste("(x3=", x3, ", ", "x4=", x4, ")", sep=""))
ggplot(emm.q1d) +
  geom_point(aes(x=Mean, y=Observation))+
  geom_errorbar(aes(y=Observation,
                    xmin=Lower, xmax=Upper),
                width = 0.25)+
  # facet_wrap(~x2*x6)+
  theme_bw()+
  xlab("Estimated Mean of Y")
```

Figure 3: Estimated marginal means for the model in Question 1.



With all other predictors besides x_3 and x_4 constant, the regression looks like:

$$E[Y_i] = C + \beta_3 I(x_{3i} = \text{Treat}) + \beta_4 I(x_{4i} = \text{High}) + \beta_9 (I(x_{3i} = \text{Treat}) \times I(x_{4i} = \text{High}))$$

When x_3 = "Control" and x_4 = "Low":

$$E[Y_i] = C$$

When x_3 = "Treat" and x_4 = "Low":

$$E[Y_i] = C + \beta_3$$

When x_3 = "Control" and x_4 = "High":

$$E[Y_i] = C + \beta_4$$

When x_3 = "Treat" and x_4 = "High":

$$E[Y_i] = C + \beta_3 + \beta_4 + \beta_9$$

All other predictors constant, when x_4 = "Low", Y_i is expected to increase by $\beta_3 = 6.51$ from Control to Treat. x_4 = "High", Y_i is expected to increase by $\beta_3 + \beta_9 = 6.51 + 15.19 = 21.70$ from Control to Treat. The effect of Treat

increases as x_4 moves from Low to High. When x_3 = "Control", Y_i is expected to increase by $\beta_4 = 3.52$ from Low to High. When x_3 = "Treat", Y_i is expected to increase by $\beta_4 + \beta_9 = 3.52 + 15.19 = 18.71$ from Low to High. The effect of High increases as x_3 moves from Control to Treat. These effects are confirmed through the table and the figure.

1.5 Part e

Untangle and interpret the $x_5 * x_6$ interaction using algebra. Use Table 2 and Figure 4 for context.

```
emm.q1e <- emtrends(mod.q1, var="x5", ~x5|x6) |>
            data.frame() |>
            rename("Slope"  = "x5.trend",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL") |>
            dplyr::select(x6, Slope, SE, df, Lower, Upper)
```

Table 2: Estimated marginal trends for the model in Question 1.

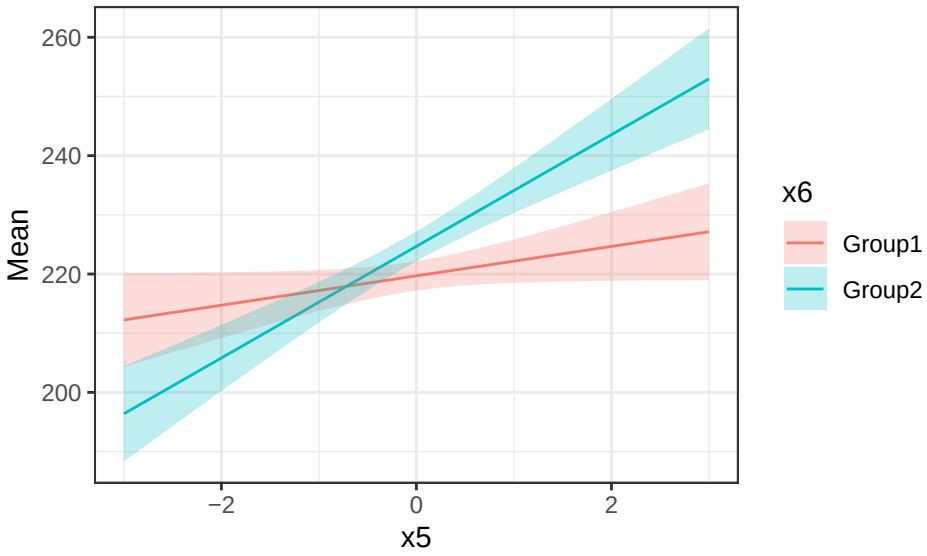
x6	Slope	SE	df	Lower	Upper
Group1	2.4835	1.2942	238	-0.0660	5.0330
Group2	9.4309	1.3367	238	6.7977	12.0642

In Figure 4, the slopes are the same and just the intercepts vary depending on x_2, x_3, and x_4.

```
emm.q1e <- emmeans(mod.q1, ~x5*x6,
                   at = list(x5 = seq(-3, 3, length.out=500))) |> # Don't average across levels: use |x2*x3*x4
  data.frame() |>
            rename("Mean"  = "emmean",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL")

ggplot(emm.q1e) +
  geom_line(aes(x=x5, y=Mean, color=x6))+
  geom_ribbon(aes(x=x5, ymin=Lower, ymax=Upper, fill=x6), alpha=0.25)+
  # facet_wrap(~x2*x3*x4)+
  theme_bw()
```

Figure 4: Estimated marginal trends for the model in Question 1.



All other predictors constant, except x_5 and x_6, the regression looks like:

$$E[Y_i] = C + \beta_5 x_5 + \beta_6 I(x_{6i} = \text{Group2}) + \beta_{10}(x_5 \times I(x_{6i} = \text{Group2}))$$

Taking the partial derivative of $E[Y_i]$ with respect to x_{5i} gives us the slope:

$$\frac{\partial E[Y_i]}{\partial x_{5i}} = \beta_5 + \beta_{10}\, I(x_{6i} = \text{Group2})$$

When $x_6 =$ "Group1", the slope of $x_5 = \beta_5 = 2.48$. When $x_6 =$ "Group2", the slope of $x_5 = \beta_5 + \beta_{10} = 2.48 + 6.95 = 9.43$. These can be seen in Table 2. Depending on the group, x_5 has less (Group1) or more (Group2) of an effect on Y.

The difference between the two groups depends on x_5. When $x_6 =$ "Group1":

$$E[Y_i] = C + \beta_5 x_{5i}$$

. When $x_6 =$ "Group2":

$$E[Y_i] = C + \beta_5 x_{5i} + \beta_6 + \beta_{10} x_{5i}$$

. We can see through Figure 2 that Group1 is higher than Group2 for low x_5, and vice versa for higher x_5.

1.6 Part f

Untangle and interpret the $x_7 * x_8$ interaction using algebra. Use Table 3 and Table 4 for context.

```
emm.q1f.x7 <- emtrends(mod.q1, var="x7", ~x7|x8,
                       at = list(x8 = mean(dat.q1$x8) + c(-1, 0 ,1)*sd(dat.q1$x8))) |>
              data.frame() |>
              rename("Slope"  = "x7.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 3: Estimated marginal trends for the model in Question 1.

x8	Slope	SE	df	Lower	Upper
8.1309	7.9230	0.6751	238	6.5932	9.2529
10.0582	9.7129	0.5015	238	8.7250	10.7008
11.9854	11.5027	0.7396	238	10.0458	12.9597

```
emm.q1f.x8 <- emtrends(mod.q1, var="x8", ~x8|x7,
                       at = list(x7 = mean(dat.q1$x7) + c(-1, 0 ,1)*sd(dat.q1$x7))) |>
              data.frame() |>
              rename("Slope"  = "x8.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 4: Estimated marginal trends for the model in Question 1.

x7	Slope	SE	df	Lower	Upper
8.3824	6.8569	0.6069	238	5.6613	8.0525
10.2208	8.5643	0.4744	238	7.6297	9.4989
12.0592	10.2716	0.7325	238	8.8286	11.7146

A Johnson Neyman analysis using the `interactions` package for `R` (Long 2024), shows that the slope of x_7 is statistically discernible for all observed values of x_8 (Figure 5), and vice versa (Figure 6).

```r
library(interactions)
johnson_neyman(mod.q1, pred = x7, modx = x8, control.fdr = T)
```

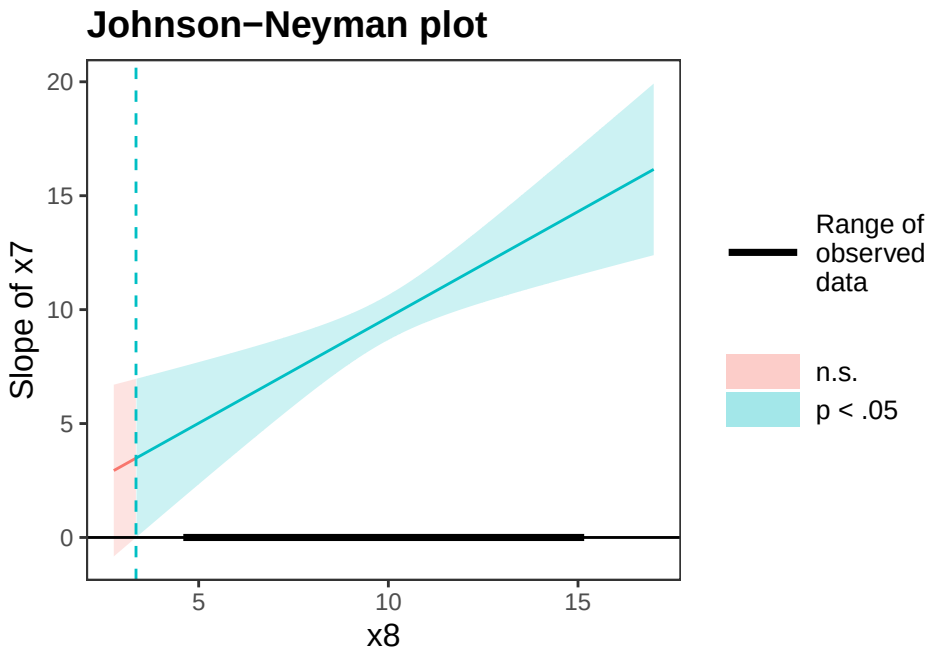
```
JOHNSON-NEYMAN INTERVAL

When x8 is OUTSIDE the interval [-13.08, 3.35], the slope of x7 is p < .05.

Note: The range of observed values of x8 is [4.69, 15.07]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 5: Johnson Neyman interval for the model in Question 1.




```r
johnson_neyman(mod.q1, pred = x8, modx = x7, control.fdr = T)
```

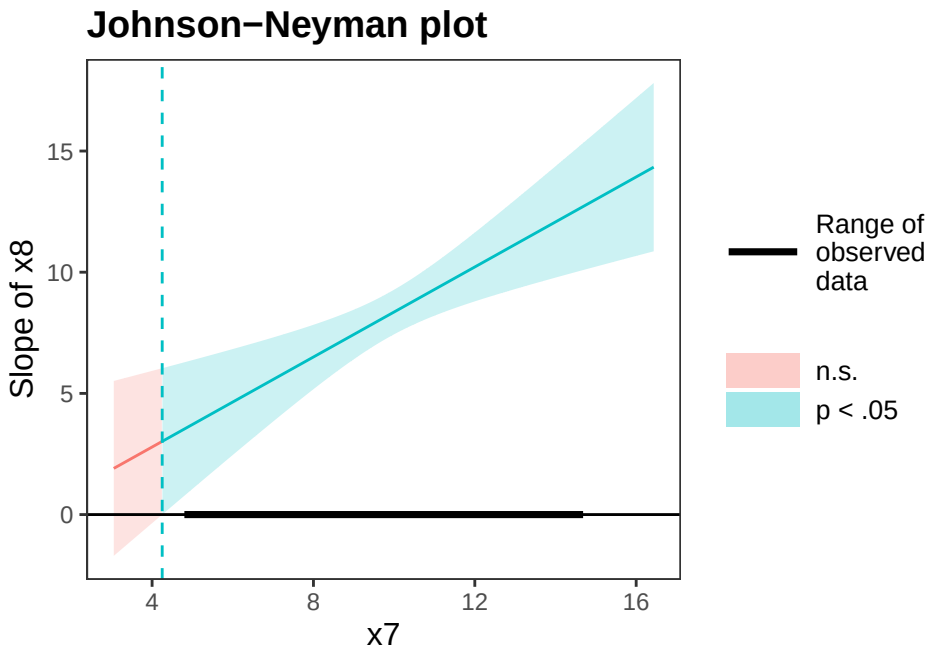
```
JOHNSON-NEYMAN INTERVAL

When x7 is OUTSIDE the interval [-9.96, 4.25], the slope of x8 is p < .05.

Note: The range of observed values of x7 is [4.89, 14.60]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 6: Johnson Neyman interval for the model in Question 1.



All other predictors constant, except x_7 and x_8, the regression looks like:

$$E[Y_i] = C + \beta_7 x_{7i} + \beta_8 x_{8i} + \beta_{11} x_{7i} x_{8i}$$

Taking the partial derivative of $E[Y_i]$ with respect to x_{7i} and x_{8i} gives us the slopes:

$$\frac{\partial E[Y_i]}{\partial x_{7i}} = \beta_7 + \beta_{11} x_{8i}$$

$$\frac{\partial E[Y_i]}{\partial x_{8i}} = \beta_8 + \beta_{11} x_{7i}$$

In words, the effect of x_7 on Y depends on x_8 and vice versa. Both slopes are linear functions of the other predictor. For each one-unit increase in x_8, the slope of x_7 changes by $\beta_{11} = 0.93$, and vice versa, seen in Tables 3 and 4. Between the first and last rows of Table 3, x_8 increases by 3.85 and the slope of x_7 increases by 3.58, giving 3.58/3.85 = 0.93.

$\beta_7 = 0.37$ represents the effect of x_7 on Y when $x_8 = 0$, and $\beta_8 = -0.93$ represents the effect of x_8 on Y when $x_7 = 0$. However, since the range of both predictors never reaches 0, these values aren't meaningful by themselves.

1.7 Part g

Standardize the continuous data and refit the regression model. Redo part f. What do you notice about the coefficient estimates and why does this make the untangling process easier?

Solution: First, we fit the regression model using `lm()`.

```
# Standardize Data
dat.z <- dat.q1 |>
  mutate(across(c(x1, x5, x7, x8), ~ as.vector(scale(.))))
# Refit Model
mod.q1z <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.z)
```

```
emm.q1g.z7 <- emtrends(mod.q1z, var="x7", ~x7|x8,
                       at = list(x8 = c(-1, 0 ,1))) |>
          data.frame() |>
          rename("Slope"  = "x7.trend",
                 "Lower" = "lower.CL",
                 "Upper" = "upper.CL") |>
          dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 5: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x8	Slope	SE	df	Lower	Upper
-1	14.57	1.24	238	12.12	17.01
0	17.86	0.92	238	16.04	19.67
1	21.15	1.36	238	18.47	23.83

```r
emm.q1g.z8 <- emtrends(mod.q1z, var="x8", ~x8|x7,
                       at = list(x7 = c(-1, 0 ,1)))|>
              data.frame() |>
              rename("Slope"  = "x8.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 6: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x7	Slope	SE	df	Lower	Upper
-1	13.22	1.17	238	10.91	15.52
0	16.51	0.91	238	14.70	18.31
1	19.80	1.41	238	17.01	22.58

AFter standardization, Taking the partial derivative of $E[Y_i]$ with respect to z_{7i} and z_{8i} gives us the slopes:

$$\frac{\partial E[Y_i]}{\partial z_{7i}} = \beta_7 + \beta_{11} z_{8i}$$

$$\frac{\partial E[Y_i]}{\partial z_{8i}} = \beta_8 + \beta_{11} z_{7i}$$

After standardizing both predictors, the values for $\beta_7, \beta_8, \beta_{11} = 17.86, 16.51, 3.29$, respectively. Now, β_7 represents the increase in Y after a one-SD increase in x_7 when x_8 is at its average ($z_8 = 0$) and β_8 represents the increase in Y after a one-SD increase x_8 when x_7 is at its average ($z_7 = 0$), due to the standardization of the predictors around 0. This makes untangling easier because we can see how the effect of each predictor behaves around the average of the other predictor, which is more informative than knowing the behavior of each predictor when the other predictor was 0.

2 Question 2

Cui and Mao (2025) investigate how freezing and cryogenic temperatures affect the strength, flexibility, stiffness, and internal breaking behavior of 3D-printed CF/PEEK composites.

Carbon fiber reinforced PEEK (CF/PEEK) is a high-performance composite used in 3D printing for aerospace, deep-sea, polar research. The researchers aimed to assess whether and how extremely cold environments impact this composite to understand how 3D-printed CF/PEEK performs in freezing conditions. This work would be important to understand its limits and develop preventative methods to protect against structural failures.

The researchers created fifteen 3D printed dog-bone-shaped test specimens and placed them in a chamber cooled with liquid nitrogen to 25°C, −25°C, −75°C, −125°C, and −175°C. To test strength of the specimen, a machine pulled both ends of the specimen until it snapped.

In the table below, the researchers record the average tensile strength for three specimens from each temperature level.

Temperature	Mean Tensile Strength
25°C	81.93 MPa
−25°C	76.63 MPa
−75°C	63.23 MPa
−125°C	32.13 MPa
−175°C	12.83 MPa

2.1 Part a

Compute the estimates we derived using calculus in lecture 16:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\sum_{i=1}^{n}(x_{1i} - \bar{x}_1)^2}.$$

```
# data from table above
dat.q2 <- tibble(
  temp     = c(25, -25, -75, -125, -175),          # x (degrees C)
  strength = c(81.93, 76.63, 63.23, 32.13, 12.83)  # y (MPa)
)

x <- dat.q2$temp
y <- dat.q2$strength

# estimate of b1 (slope) using calculus
b1.hat <- sum((x - mean(x)) * (y - mean(y))) / sum((x - mean(x))^2)

# estimate of b0 (intercept) using calculus
b0.hat <- mean(y) - b1.hat * mean(x)

b0.hat # intercept
```

```
[1] 80.755
```

```
b1.hat # slope
```

```
[1] 0.3654
```

2.2 Part b

Compute the estimates we derived using linear algebra in lecture 16:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

Hint: Run `?matrix()` in `R` for a quick reminder about how to create a matrix.

```
# remake data from table above
dat.q2 <- tibble(
  temp     = c(25, -25, -75, -125, -175),          # x (degrees C)
  strength = c(81.93, 76.63, 63.23, 32.13, 12.83)  # y (MPa)
)

x <- dat.q2$temp
```

```r
y <- dat.q2$strength

# make matrices X and Y
x.mat <- cbind(1, x)
y.mat <- matrix(data = y)

# compute estimates using linear algebra
estimates <- solve(t(x.mat) %*% x.mat) %*% t(x.mat) %*% y.mat
b0.hat.b <- estimates[1]
b1.hat.b <- estimates[2]

b0.hat.b
```

```
[1] 80.755
```

```r
b1.hat.b
```

```
[1] 0.3654
```

2.3 Part c

Compute the maximum likelihood estimates using `optim()` in `R` as described in lecture 16:

$$\max_{\beta_0,\beta_1,\sigma} \mathcal{L}(\theta = \{\beta_0, \beta_1, \sigma\}).$$

```r
# Negative log-likelihood where theta = c(beta0, beta1, sigma)
neg.llike <- function(theta, x, y) {
  b0 <-    theta[1]
  b1 <-    theta[2]
  sigma <- theta[3]
  mu <- b0 + b1 * x                # E(Y | x)
  -sum(dnorm(y, mean = mu, sd = sigma, log = TRUE))
}

mle <- optim(par = c(mean(y), 0, sd(y)),   # starting values
             fn  = neg.llike,
             x = x, y = y,
             control = list(maxit = 10000, reltol = 1e-12))

mle # ($par)
```

```
$par
[1] 80.754999  0.365400  6.490294

$value
[1] 16.44622

$counts
function gradient
     190       NA

$convergence
[1] 0

$message
NULL
```

2.4 Part d

Fit the model using `lm()` and plot the resulting model. Compare to the results from parts (a) through (c).

```
mod.q2 <- lm(strength ~ temp, data=dat.q2)
summary(mod.q2)
```

```
Call:
lm(formula = strength ~ temp, data = dat.q2)

Residuals:
    1     2     3     4     5
-7.96  5.01  9.88 -2.95 -3.98

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 80.75500    5.46239  14.784 0.000671 ***
temp         0.36540    0.05299   6.895 0.006250 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.379 on 3 degrees of freedom
Multiple R-squared:  0.9406,    Adjusted R-squared:  0.9209
F-statistic: 47.54 on 1 and 3 DF,  p-value: 0.00625
```

The results match the estimates found through parts (a) – (c). $\hat{\beta}_0 = 80.76$ (Intercept), and $\hat{\beta}_1 = 0.365$ (slope). In words, the strength at 0 degrees Celsius is 80.76 MPa and the strength will decrease by 0.365 MPa with every one-degree decrease in temperature.

3 Question 3

Bracht et al. (2016) evaluate MFAP4 as a biomarker for severity of fibrosis in hepatitic C patients.

```
dat.mfap4 <- read_csv("Bracht2016.csv")
```

3.1 Part a

Fit the model predicting `MFAP4` using `Fibrosis.Stage`. Write out the resulting estimate model and interpret the coefficients.

```
mod.q3a <- lm(MFAP4 ~ Fibrosis.Stage, data = dat.mfap4)
summary(mod.q3a)
```

```
Call:
lm(formula = MFAP4 ~ Fibrosis.Stage, data = dat.mfap4)

Residuals:
    Min      1Q  Median      3Q     Max
-20.318  -5.518  -2.651   2.070  81.816

Coefficients:
```

```
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)             9.230      1.102   8.378 4.73e-16 ***
Fibrosis.StageStage 1   1.301      1.372   0.948   0.3433
Fibrosis.StageStage 2   3.154      1.444   2.184   0.0294 *
Fibrosis.StageStage 3  11.742      1.724   6.812 2.59e-11 ***
Fibrosis.StageStage 4  15.388      1.724   8.928  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.85 on 537 degrees of freedom
Multiple R-squared:  0.1965,    Adjusted R-squared:  0.1905
F-statistic: 32.84 on 4 and 537 DF,  p-value: < 2.2e-16
```

The resulting regression model:

$\hat{MFAP4_i} = 9.23 + 1.31 I(Fibrosis.Stage_i = \text{Stage 1}) + 3.15 I(Fibrosis.Stage_i = \text{Stage 2}) + 11.72 I(Fibrosis.Stage_i = \text{Stage 3}) + 15.$

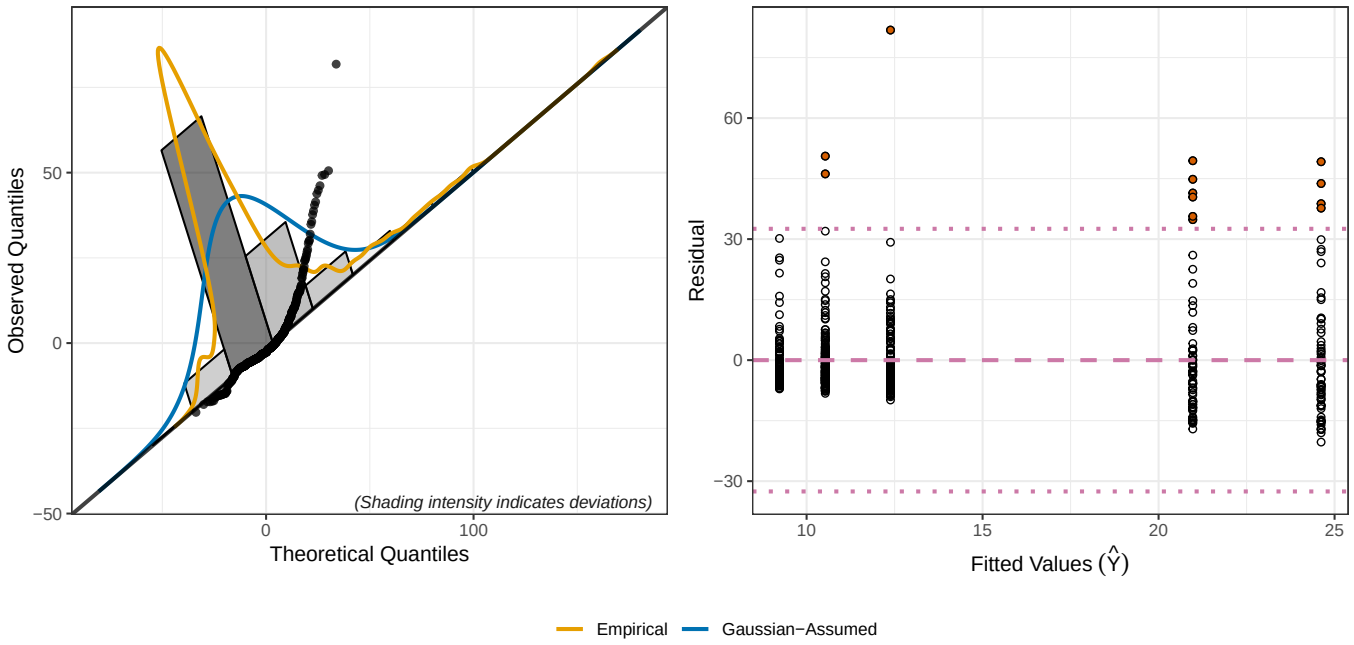
9.23 represents $E[MFAP4_i]$ at Stage 0. 1.31 represents the increase in $E[MFAP4_i]$ from Stage 0 to Stage 1. 3.15 represents the increase in $E[MFAP4_i]$ from Stage 0 to Stage 2. 11.74 represents the increase in $E[MFAP4_i]$ from Stage 0 to Stage 3. 15.39 represents the increase in $E[MFAP4_i]$ from Stage 0 to Stage 4.

3.2 Part b

Conduct a residual analysis on the model in part a to demonstrate how to identify when you need a log transformation.

```
1 plotResiduals(mod.q3a)
```

Figure 7: Residual diagnostics for the Question 3 model.



The residuals do not appear to be Gaussian distributed, the values move away from the qq-plot as they increase. Many of the residuals are clusted around 0 or below, with only positive outliers present. This spread is conducive to a log transformation.

3.3 Part c

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

```
mod.q3log <- lm(log(MFAP4) ~ Fibrosis.Stage, data = dat.mfap4)
summary(mod.q3log)
```

```
Call:
lm(formula = log(MFAP4) ~ Fibrosis.Stage, data = dat.mfap4)

Residuals:
     Min       1Q   Median       3Q      Max
-1.55668 -0.41861 -0.04282  0.35845  2.21571

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)            2.02572    0.06178  32.789  < 2e-16 ***
Fibrosis.StageStage 1  0.13336    0.07694   1.733 0.083637 .
Fibrosis.StageStage 2  0.30399    0.08099   3.753 0.000193 ***
Fibrosis.StageStage 3  0.77408    0.09666   8.009 7.24e-15 ***
Fibrosis.StageStage 4  0.98957    0.09666  10.238  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6085 on 537 degrees of freedom
Multiple R-squared:  0.2312,    Adjusted R-squared:  0.2255
F-statistic: 40.38 on 4 and 537 DF,  p-value: < 2.2e-16
```

The resulting regression model:

$\hat{MFAP4_i} = 2.03 + 0.13 I(Fibrosis.Stage_i = \text{Stage 1}) + 0.30 I(Fibrosis.Stage_i = \text{Stage 2}) + 0.77 I(Fibrosis.Stage_i = \text{Stage 3}) + 0.99$

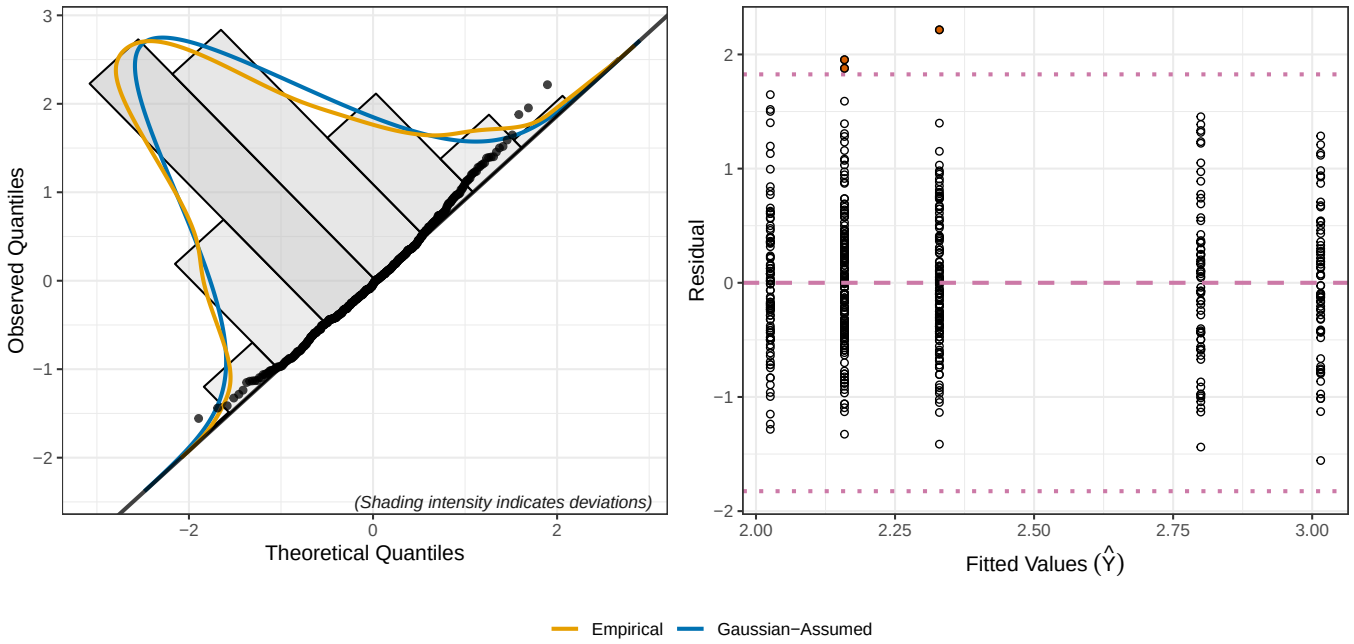
2.03 represents $E[ln(MFAP4_i)]$ at Stage 0. 0.13 represents the increase in $E[ln(MFAP4_i)]$ from Stage 0 to Stage 1. 0.30 represents the increase in $E[ln(MFAP4_i)]$ from Stage 0 to Stage 2. 0.77 represents the increase in $E[ln(MFAP4_i)]$ from Stage 0 to Stage 3. 0.99 represents the increase in $E[ln(MFAP4_i)]$ from Stage 0 to Stage 4.

3.4 Part d

Conduct a residual analysis on the model in part c to demonstrate that the conditions for using regression are met.

```r
plotResiduals(mod.q3log)
```

Figure 8: Residual diagnostics for the Question 3 model after log transformation.



The residuals seem to be Gaussian distributed, and the residuals are more evenly centered around 0, with very few outliers. This is a better fit for a linear regression than the raw MFAP4 counts.

3.5 Part e

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`, `Age`, and `Gender`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

```r
mod.q3e <- lm(log(MFAP4) ~ Fibrosis.Stage + Age + Gender, data = dat.mfap4)
summary(mod.q3e)
```

```
Call:
lm(formula = log(MFAP4) ~ Fibrosis.Stage + Age + Gender, data = dat.mfap4)

Residuals:
     Min       1Q   Median       3Q      Max
-1.55799 -0.40010 -0.03682  0.35968  2.28903

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)           1.518667   0.115808  13.114  < 2e-16 ***
Fibrosis.StageStage 1 0.126858   0.073594   1.724  0.08533 .
Fibrosis.StageStage 2 0.212539   0.079095   2.687  0.00743 **
Fibrosis.StageStage 3 0.598866   0.096575   6.201 1.12e-09 ***
Fibrosis.StageStage 4 0.828547   0.097208   8.523  < 2e-16 ***
Age                   0.013110   0.002123   6.175 1.31e-09 ***
Gendermale           -0.143405   0.050807  -2.823  0.00494 **
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5812 on 535 degrees of freedom
Multiple R-squared:  0.3012,    Adjusted R-squared:  0.2934
F-statistic: 38.44 on 6 and 535 DF,  p-value: < 2.2e-16
```

The resulting regression model:

$\widehat{MFAP4}_i = 1.52 + I(Fibrosis.Stage_i = \text{Stage 1}) + 0.30 I(Fibrosis.Stage_i = \text{Stage 2}) + 0.77 I(Fibrosis.Stage_i = \text{Stage 3}) + 0.99 I(F$

2.03 represents $E[ln(MFAP4_i)]$ at Stage 0. 0.13 represents the increase in $E[ln(MFAP4_i)]$ from Stage 0 to Stage 1. 0.21 represents the increase in $E[ln(MFAP4_i)]$ from Stage 0 to Stage 2. 0.60 represents the increase in $E[ln(MFAP4_i)]$ from Stage 0 to Stage 3. 0.83 represents the increase in $E[ln(MFAP4_i)]$ from Stage 0 to Stage 4. 0.01 represents the change in $E[ln(MFAP4_i)]$ after a one-year increase in age. -0.14 represents the change in $E[ln(MFAP4)]$ from female to male patients.

4 Question 4

4.1 Part a

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Gaussian}(\mu = 0, \sigma = 1) \\
y_i &= 2 + 1.5 x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

```r
set.seed(1206)
R <- 1000
n <- 30

# create a space for the estimates
estimates_a <- tibble(b0 = rep(NA, R),
                      b1 = rep(NA, R))

# simulate the data
for (r in 1:R) {
  # X ~ Uniform(a = 0, b = 10)
  dat.q4 <- tibble(x = runif(n, min = 0, max = 10)) |>
    mutate(y = 2             # Intercept
           + 1.5 * x         # Xi predictor
           + rnorm(n, 0, 1)) # Gaussian distributed errors

  # model
  mod.q4 <- lm(y ~ x, data = dat.q4)

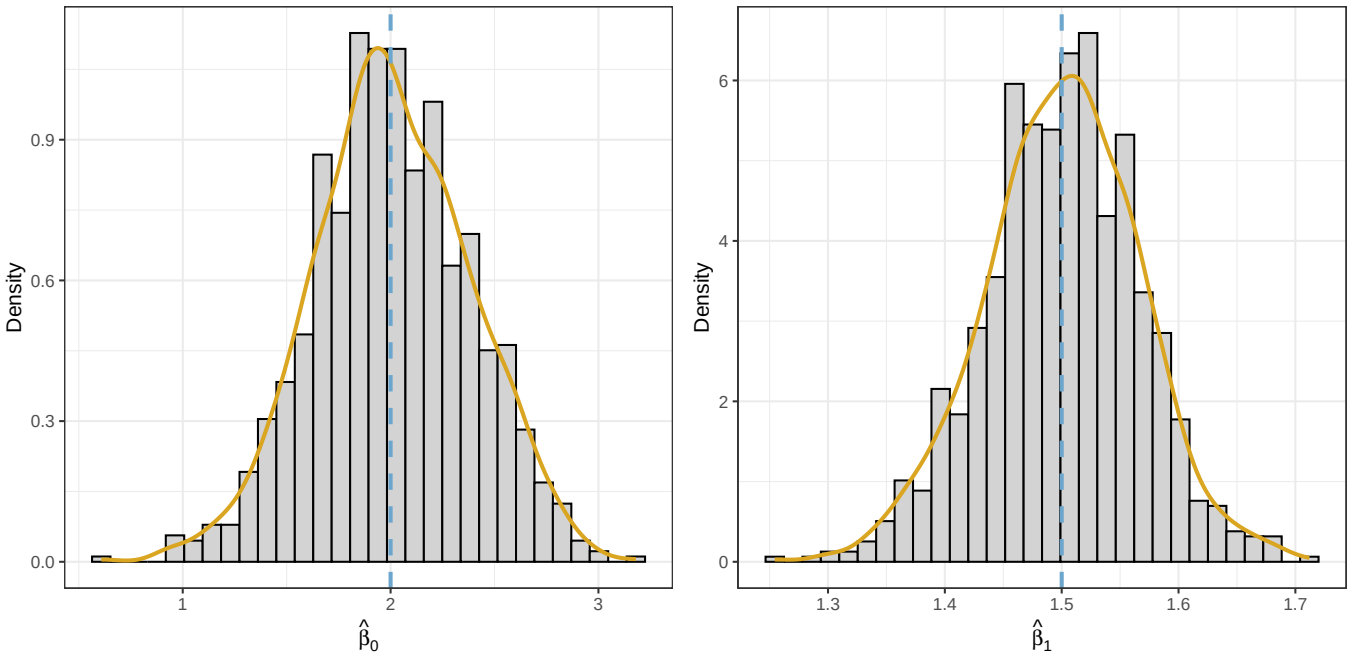
  # store estimates
  estimates_a$b0[r] <- coef(mod.q4)[1] # b0
  estimates_a$b1[r] <- coef(mod.q4)[2] # b1
}

```

```r
# plot the estimates
p.b0.a <- ggplot(estimates_a, aes(x = b0)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "lightgrey", color = "black") +
  geom_density(color = "goldenrod", linewidth = 1) +
  geom_vline(xintercept = 2, color = "skyblue3",
             linetype = "dashed", linewidth = 1) +       # true beta_0
  labs(x = expression(hat(beta)[0]), y = "Density") +
  theme_bw()

p.b1.a <- ggplot(estimates_a, aes(x = b1)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "lightgrey", color = "black") +
  geom_density(color = "goldenrod", linewidth = 1) +
  geom_vline(xintercept = 1.5, color = "skyblue3",
             linetype = "dashed", linewidth = 1) +       # true beta_1
  labs(x = expression(hat(beta)[1]), y = "Density") +
  theme_bw()

p.b0.a | p.b1.a
```



The values for $\hat{\beta}_0$ and $\hat{\beta}_1$ are approximately normally distributed around their true values of 2 and 1.5, respectively. The spread of values for $\hat{\beta}_0$ is larger than the spread of values around $\hat{\beta}_1$.

4.2 Part b

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Shifted Exponential}(\lambda = 1, \mu = 0) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

Hint: Use the `tolerance::r2exp()` function with `rate=1` and `shift=-1`.

```r
set.seed(1206)
R <- 1000
n <- 30

# create a space for the estimates
estimates_b <- tibble(b0 = rep(NA, R),
                      b1 = rep(NA, R))

# simulate the data
for (r in 1:R) {
  # X ~ Uniform(a = 0, b = 10)
  dat.q4 <- tibble(x = runif(n, min = 0, max = 10)) |>
    mutate(y = 2             # Intercept
           + 1.5 * x         # Xi predictor
           # Shifted exponentially distributed errors
           + r2exp(n, rate = 1, shift = -1))

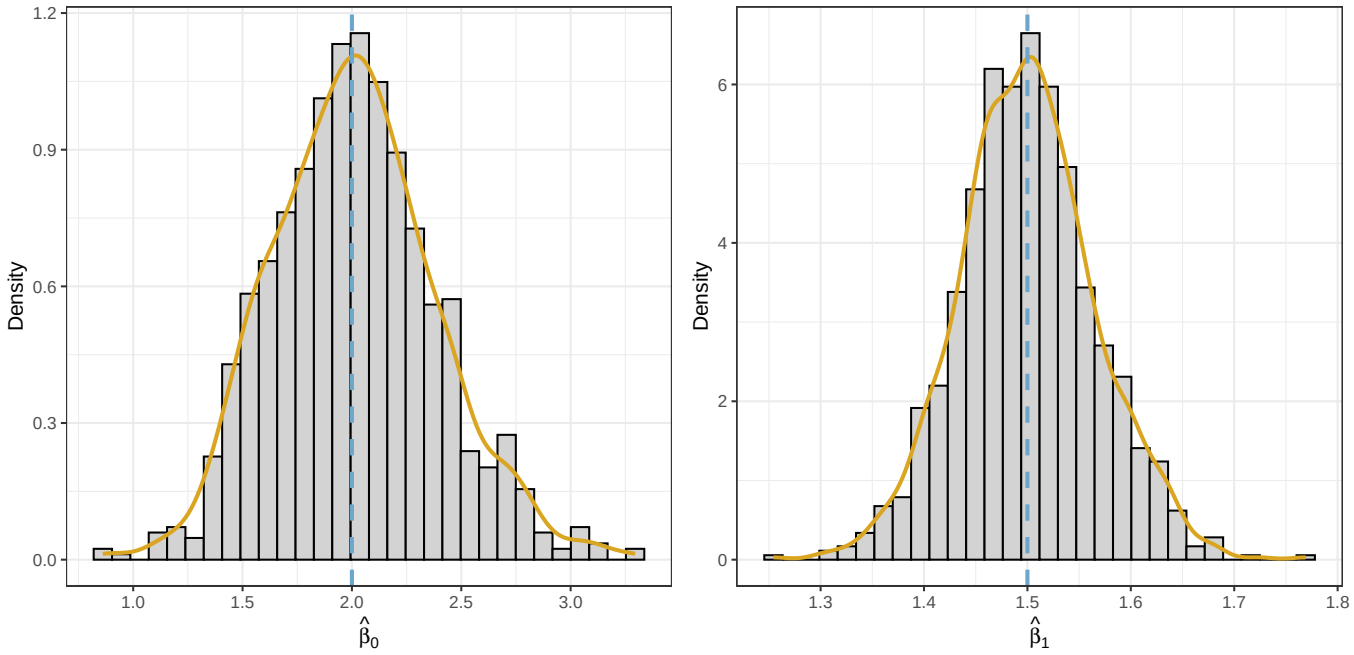
  # model
  mod.q4 <- lm(y ~ x, data = dat.q4)

  # store estimates
  estimates_b$b0[r] <- coef(mod.q4)[1] # b0
  estimates_b$b1[r] <- coef(mod.q4)[2] # b1
}

# plot the estimates
p.b0.b <- ggplot(estimates_b, aes(x = b0)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "lightgrey", color = "black") +
  geom_density(color = "goldenrod", linewidth = 1) +
  geom_vline(xintercept = 2, color = "skyblue3",
             linetype = "dashed", linewidth = 1) +       # true beta_0
  labs(x = expression(hat(beta)[0]), y = "Density") +
  theme_bw()

p.b1.b <- ggplot(estimates_b, aes(x = b1)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "lightgrey", color = "black") +
  geom_density(color = "goldenrod", linewidth = 1) +
  geom_vline(xintercept = 1.5, color = "skyblue3",
             linetype = "dashed", linewidth = 1) +       # true beta_1
  labs(x = expression(hat(beta)[1]), y = "Density") +
  theme_bw()

p.b0.b | p.b1.b
```



The values for $\hat{\beta}_0$ and $\hat{\beta}_1$ are approximately normally distributed around their true values of 2 and 1.5, respectively. Again, the spread of values for $\hat{\beta}_0$ is larger than the spread of values around $\hat{\beta}_1$. The plots look very similar to the plots from part a.

4.3 Part c

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Cauchy}(m = 0, s = 1) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

```
set.seed(1206)
R <- 1000
n <- 30

# create a space for the estimates
estimates_c <- tibble(b0 = rep(NA, R),
                      b1 = rep(NA, R))

# simulate the data
for (r in 1:R) {
  # X ~ Uniform(a = 0, b = 10)
  dat.q4 <- tibble(x = runif(n, min = 0, max = 10)) |>
    mutate(y = 2              # Intercept
           + 1.5 * x          # Xi predictor
           # Cauchy distributed errors
           + rcauchy(n, location = 0, scale = 1))

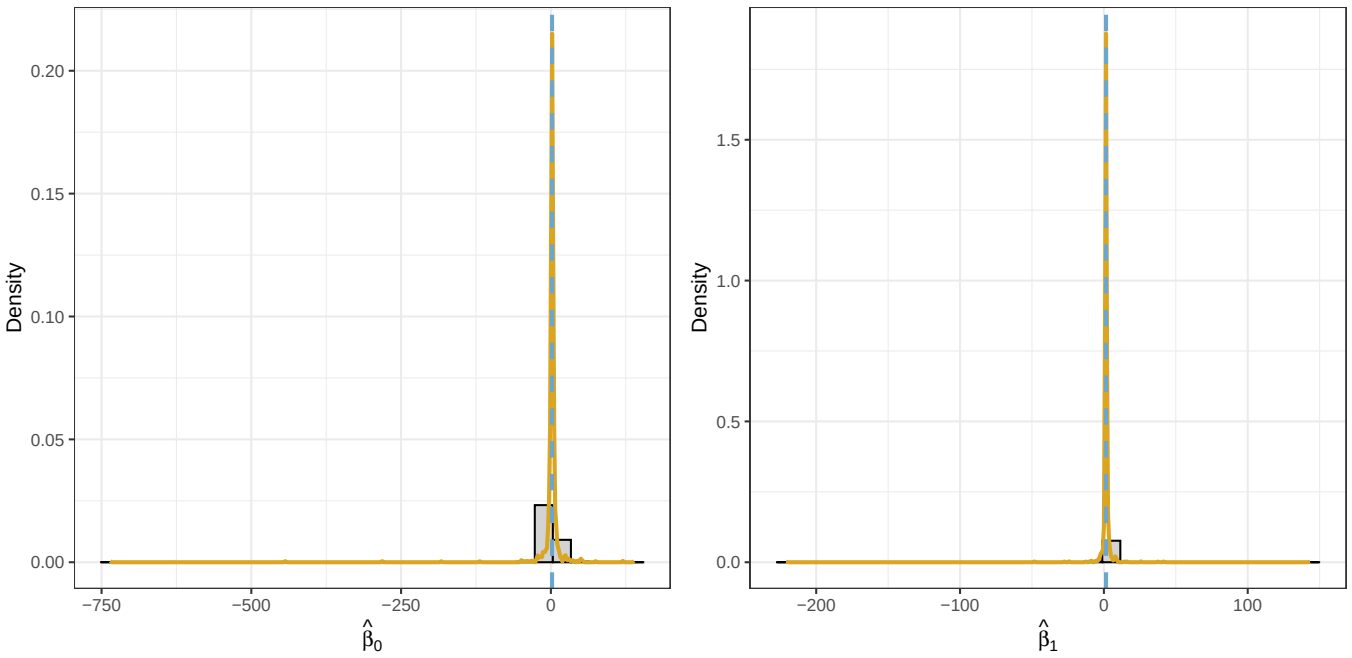
  # model
  mod.q4 <- lm(y ~ x, data = dat.q4)
```

```r
  # store estimates
  estimates_c$b0[r] <- coef(mod.q4)[1] # b0
  estimates_c$b1[r] <- coef(mod.q4)[2] # b1
}

# plot the estimates
p.b0.c <- ggplot(estimates_c, aes(x = b0)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "lightgrey", color = "black") +
  geom_density(color = "goldenrod", linewidth = 1) +
  geom_vline(xintercept = 2, color = "skyblue3",
             linetype = "dashed", linewidth = 1) +       # true beta_0
  labs(x = expression(hat(beta)[0]), y = "Density") +
  theme_bw()

p.b1.c <- ggplot(estimates_c, aes(x = b1)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "lightgrey", color = "black") +
  geom_density(color = "goldenrod", linewidth = 1) +
  geom_vline(xintercept = 1.5, color = "skyblue3",
             linetype = "dashed", linewidth = 1) +       # true beta_1
  labs(x = expression(hat(beta)[1]), y = "Density") +
  theme_bw()

p.b0.c | p.b1.c
```



The values for $\hat{\beta}_0$ and $\hat{\beta}_1$ not normally distributed around their true values of 2 and 1.5, respectively. While most of the estimates seem to center around their true values, it seems like the estimates that are different vary by a lot more than the 2 scenarios above.

References

Bracht, Thilo, Christian Mölleken, Maike Ahrens, et al. 2016. “Evaluation of the Biomarker Candidate MFAP4 for

Non-Invasive Assessment of Hepatic Fibrosis in Hepatitis C Patients." *Journal of Translational Medicine* 14 (1): 201.

Cui, Mengyang, and Jian Mao. 2025. "Effect of Low Temperature on Mechanical Properties and Fracture Behavior of 3D Printed Short Carbon Fiber Reinforced Polyether Ether Ketone Composites (CF/PEEK)." *Scientific Reports* 15 (1): 36244.

Long, Jacob A. 2024. *Interactions: Comprehensive, User-Friendly Toolkit for Probing Interactions*. https://doi.org/10.32614/CRAN.package.interactions.